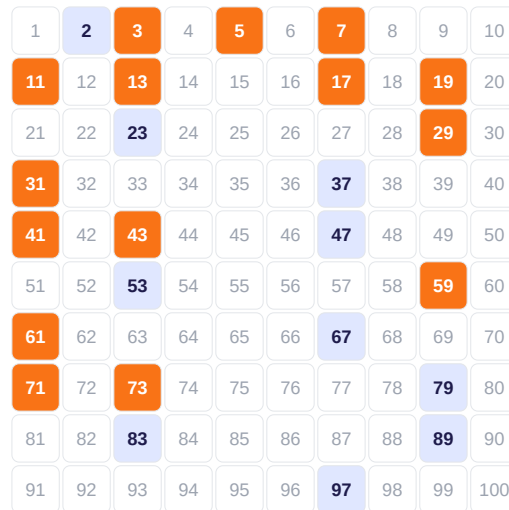


Twin Prime Conjecture

Primes that are just 2 apart, like 11 and 13, keep showing up forever.



Primes up to 100 are shaded. Twin pairs are orange. They get rarer, but do they ever stop?

TRY IT

3 and 5

5 and 7

11 and 13

17 and 19

29 and 31

41 and 43

59 and 61

71 and 73

WHAT MATHEMATICIANS KNOW

- Euclid proved over 2,000 years ago that there are infinitely many primes. Whether there are infinitely many twins is still unknown.
- In 2013 Yitang Zhang proved that some gap under 70 million repeats forever. A worldwide collaboration soon brought it down to 246.
- So primes keep coming within 246 of each other forever. Exactly 2 apart is the open question.

WHY IT'S STILL UNSOLVED

Primes thin out as numbers grow, so twins get rarer. The evidence says they never run out, but getting from “some gap of at most 246” down to a gap of exactly 2 is the step no one knows how to take.

TRY THIS WITH STUDENTS

- Shade the primes on a hundred chart, then circle the twin pairs. Where do twins sit?
- Every twin pair after 3 and 5 has a multiple of 6 between them (6, 12, 18, 30...). Ask students to notice it, then explain why it must happen.
- Ask: Can three numbers spaced by 2, like 3, 5, 7, ever all be prime again? (No: one of them is always a multiple of 3.)

WORDS TO KNOW

Twin primes two primes that differ by 2

Infinitely many however many you list, there is always another